

Fig. 3: Circuit diagram of the PIN diode based fire sensor

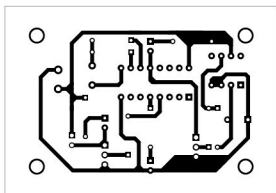


Fig. 4: PCB layout of the PIN diode fire alarm

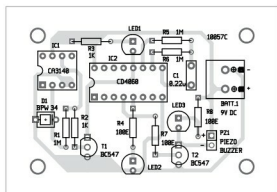


Fig. 5: Component layout of the PCB

Large-value feedback resistor R1 sets the gain of the transimpedance amplifier since it is in inverting configuration. Connection of non-inverting input to ground provides low impedance load for the photodiode,

which keeps the photodiode voltage low.

The photodiode operates in the photo-voltaic mode with no external bias. Feedback of the op-amp keeps the photodiode current equal to the feedback current through R1. So the input offset voltage due to the photodiode is very low in this self-biased photovoltaic mode. This permits a large gain without any large-output offset voltage. This configuration is selected to get large gain in low-light conditions.

Normally, in ambient light condition, photocurrent from the PIN diode is very low; it keeps output of IC1 low. When the PIN diode detects visible light or IR from fire, its photo current increases and transimpedance ampli-

fier IC1 converts this current to corresponding output voltage. High output from IC1 activates transistor T1 and LED1 glows. This indicates that the circuit has detected fire. When T1 conducts, it takes reset pin

12 of IC2 to ground potential and CD4060 starts oscillating.

IC2 is a binary counter with ten outputs that turn high one by one when it oscillates due to C1 and R6. Oscillation of IC2 is indicated by the blinking of LED2. When output Q6 (pin 4) of IC2 turns high after 15 seconds, T2 conducts and activates piezo buzzer PZ1, and LED3 also glows. The alarm repeats again after 15 seconds if

fire persists.

You can also turn on an AC alarm that produces a loud sound by replacing PZ1 with a relay circuitry (not shown here). The AC alarm is activated through contacts of the relay used for this purpose.

Construction and testing

An actual-size, single-side PCB for the PIN diode based fire sensor is shown in Fig. 4 and its component layout in Fig. 5. Enclose the PCB in a small box in such a way that you can connect PIN diode BPW34 easily at the rear side of the box. Install the PIN diode in a suitable place and cover it such that normal light/sunlight does not fall on it.

Testing the circuit is simple. Normally, when there is no fire flame near the PIN diode, the piezo buzzer does not sound. When a fire flame is sensed by the PIN diode, piezo buzzer sounds an alarm. Its detection range is around two metres. It can also detect sparks in the mains wiring due to short-circuit. **BY**

